

DSA Day 21: Non-recursive traversal & tree building

Book: Data Structures Through C in Depth - Ch6 §6.9.1-6.11 · pp 188-201 | Code in Python

Overview

Iterative traversals use an explicit stack. A tree is uniquely rebuilt from inorder + one of pre/postorder.

Key concepts to master

- Explicit stack traversal
- Rebuild from inorder+preorder
- Height, expression tree

Python implementation

```
def inorder_iter(root):
    st=[]; cur=root
    while st or cur:
        while cur: st.append(cur); cur=cur.left
        cur=st.pop(); print(cur.data); cur=cur.right
```

Complexity

$O(n)$ time, $O(h)$ space.

Common pitfalls

- Not pushing the whole left spine
- Assuming preorder alone rebuilds a tree

Practice

- Iterative inorder
- Rebuild from inorder+preorder

